PULLBACK OF SHEAVES OF MODULES IS LEFT ADJOINT TO PUSHFORWARD OF RINGED SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sheaves.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Tensor-restrict adjunction for sheaves of modules: Hom_{O_X}(O_X tensor_{f^{-1}O_Y} f^{-1}G, F) = Hom_{f^{-1}O_Y}(f^{-1}G, F_{f^{-1}O_Y}). invoked at proof lines 3391-3400. Provides the first equality of the chain.
- Push-pull adjunction for modules: Mor_{Mod(f^{-1}O_Y)}(f^{-1}G, F) = Hom_{O_Y}(G, f_*F). invoked at proof lines 3400-3402. Provides the second equality of the chain.

Source reference. Original lemma 008A: sources/stacks-sheaves/pinned-original-source.tex, source lines 2133–2133. Cited at proof lines 3400-3401 as the justification of the first equality; load-bearing inside the counted span.

Source reference. Original lemma 008Y: sources/stacks-sheaves/pinned-original-source.tex, source lines 3150–3150. Cited at proof lines 3401-3402 as the justification of the second equality; load-bearing inside the counted span.

Lemma 0.2. *Let $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_Y$-modules. There is a canonical bijection*

$$\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F}).$$

In other words: the functor f^* is the left adjoint to f_*.

Proof. This follows from the work we did before:

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) &= \mathrm{Mor}_{Mod(\mathcal{O}_X)}(\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{G}, \mathcal{F}) \\ &= \mathrm{Mor}_{Mod(f^{-1}\mathcal{O}_Y)}(f^{-1}\mathcal{G}, \mathcal{F}_{f^{-1}\mathcal{O}_Y}) \\ &= \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F}). \end{aligned}$$

Here we use Lemmas 008A and 008Y. □

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